Week 9 — Activity: Half-Space Cooling of the Oceanic Lithosphere

Geophysics 3A: Potential Fields & Geothermics

May 1, 2026

Purpose

To develop physical intuition for transient cooling of the oceanic lithosphere using the half-space cooling model.

Model Assumptions

- The mantle is initially at uniform temperature.
- Surface temperature is suddenly reduced and held fixed.
- No fixed lithospheric thickness is imposed.

Solution Form

The temperature evolution is given by

$$T(z,t) = T_m \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Tasks

1. Sketch temperature–depth profiles for increasing times.
2. Identify the thermal boundary layer and describe how its thickness evolves.
3. Estimate how the depth to a given isotherm scales with seafloor age.

Geological Interpretation

- Oceanic lithosphere thickens thermally with age.
- Heat flow decreases as the plate cools away from mid-ocean ridges.

Key Insight

Explain why the half-space cooling model predicts unbounded lithospheric thickening.